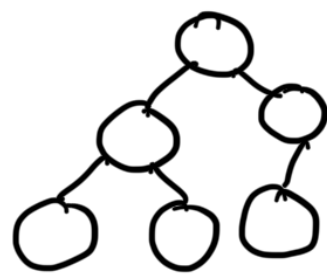
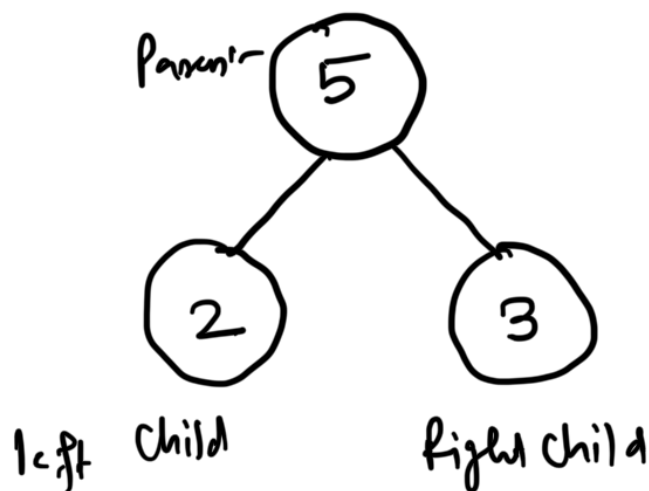


Heap : Data structure : Complete Binary Tree

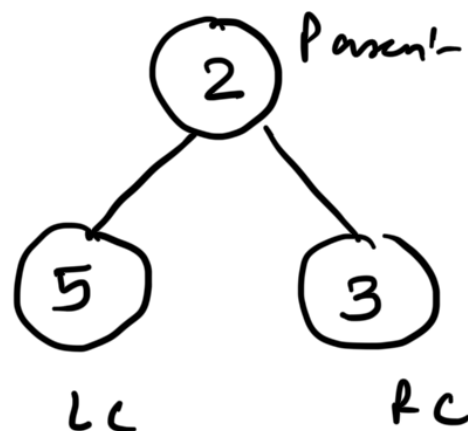
- A special form of Complete BT that key value of each node is smaller (larger) than the key value of its children.



almost $\{0, 1, 2\}$
left child 1st pref



Max Heap : Parent > largest



Min Heap

Parent < smallest

Max-Heap :- Root node has the largest key

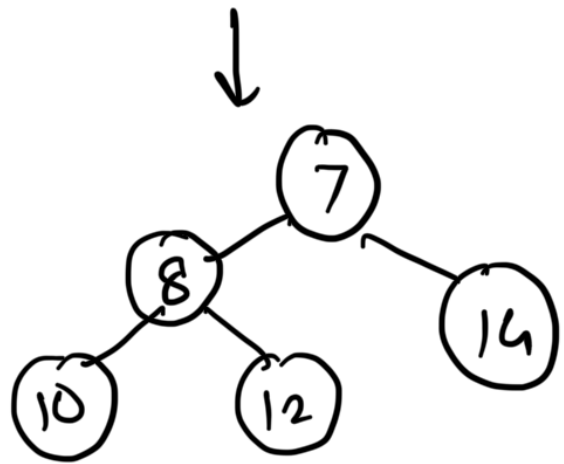
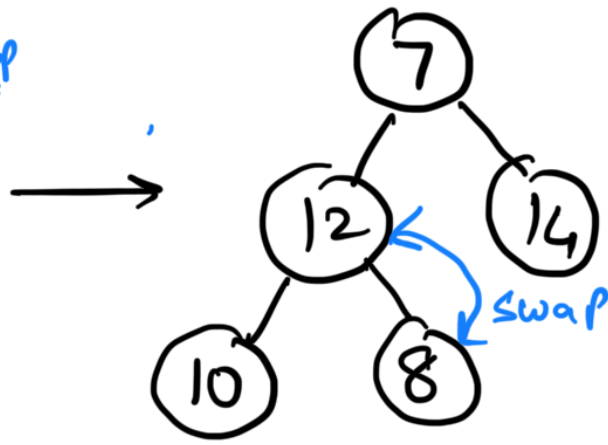
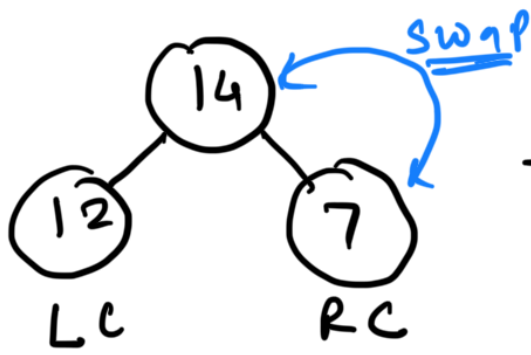
A max-heap is a tree in which the key value in each node is smaller than the key value in the root node

Min-Heap :- Root node has the smallest key

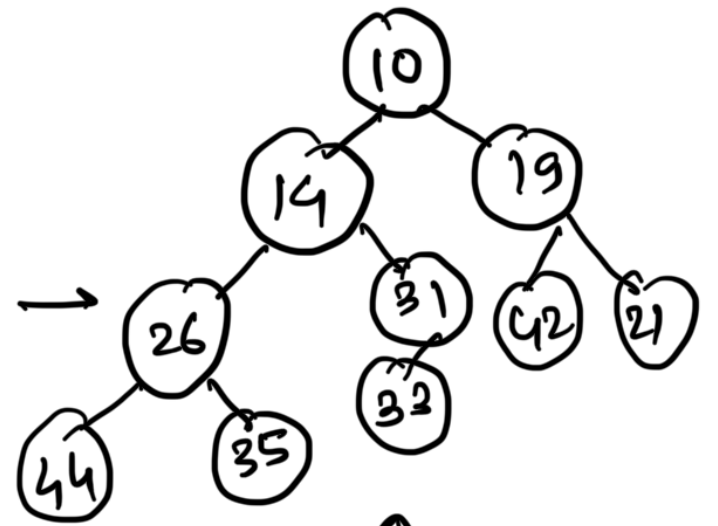
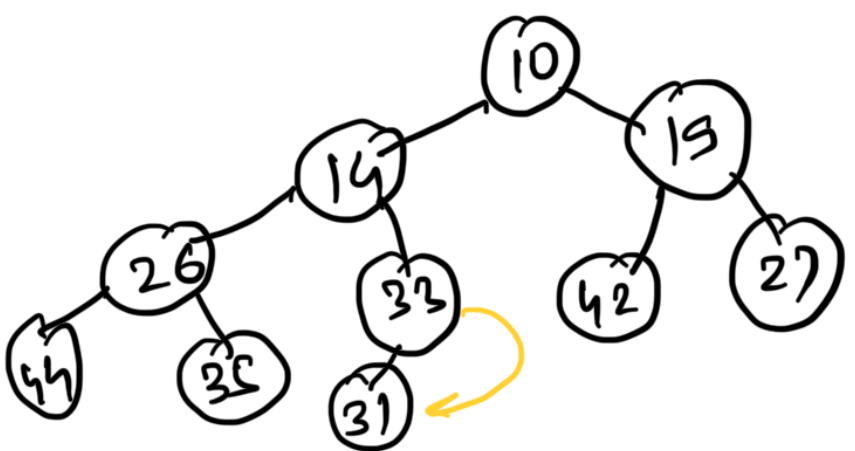
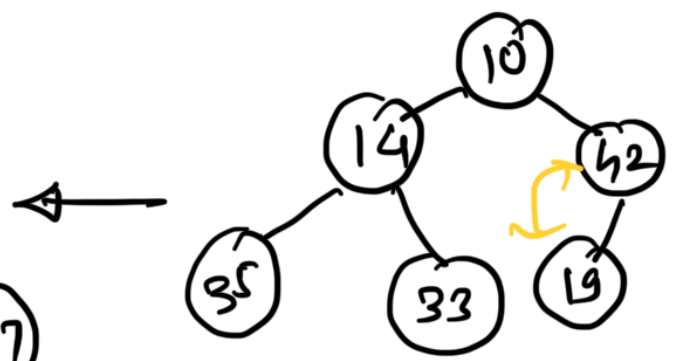
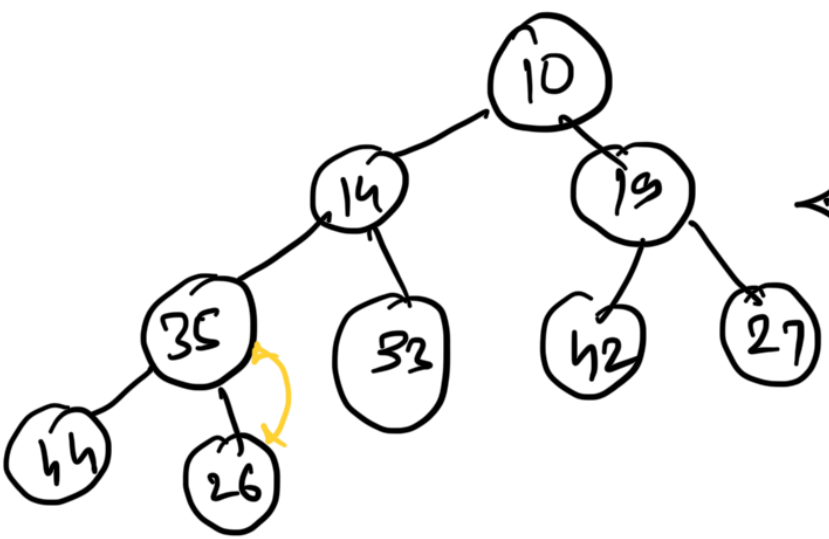
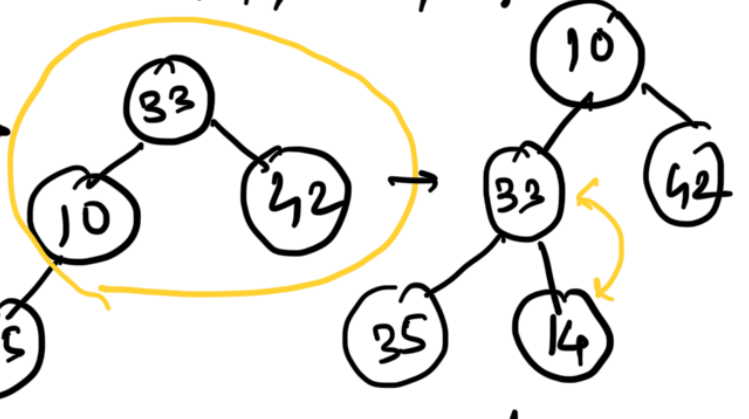
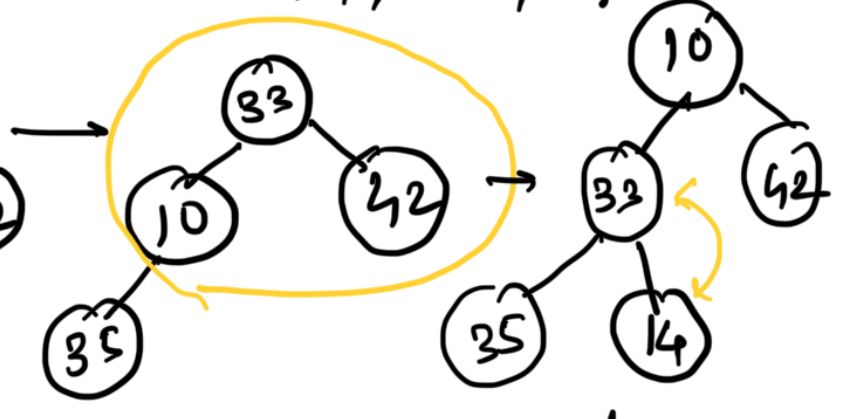
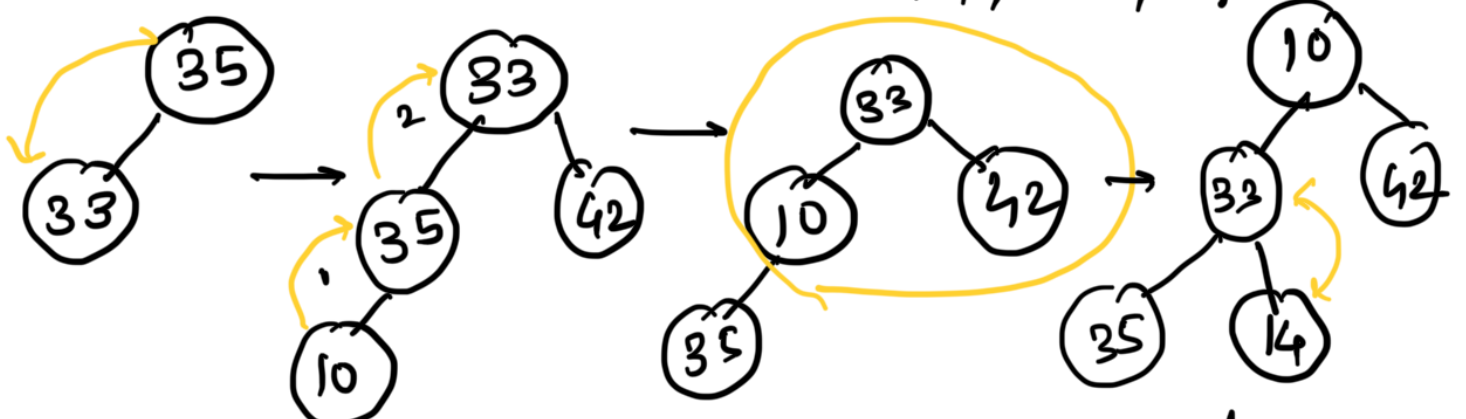
A min-heap is a tree in which the key value in each node is greater than the key value in the root node

Example : 14, 12, 7, 10, 8

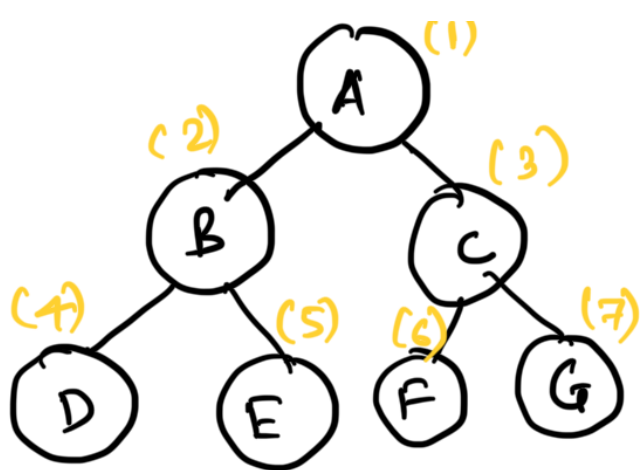




Insertion : Min Heap : 35, 33, 42, 10, 14, 19, 27, 44, 26, 31



Final Answer
Min Heap



If Node is at index i

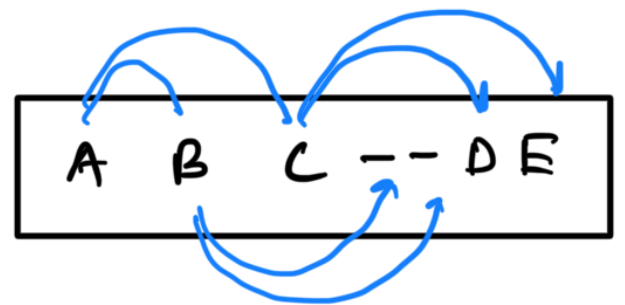
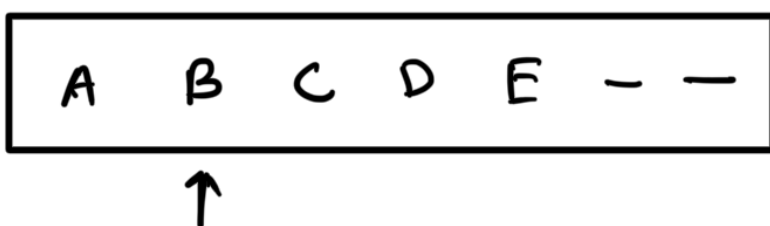
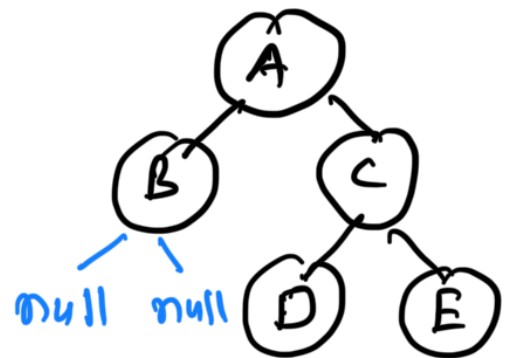
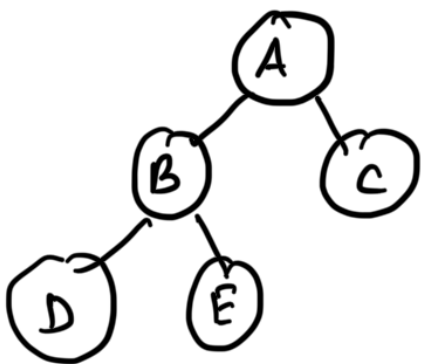
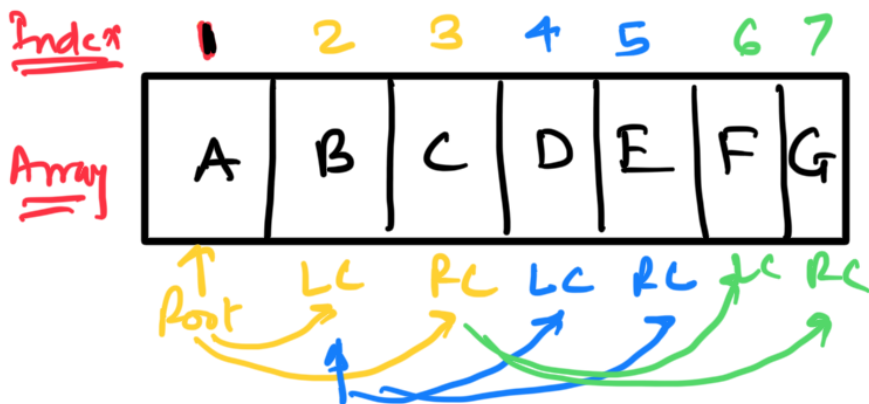
$$LC \rightarrow 2 \times i$$

$$RC \rightarrow 2 \times i + 1$$

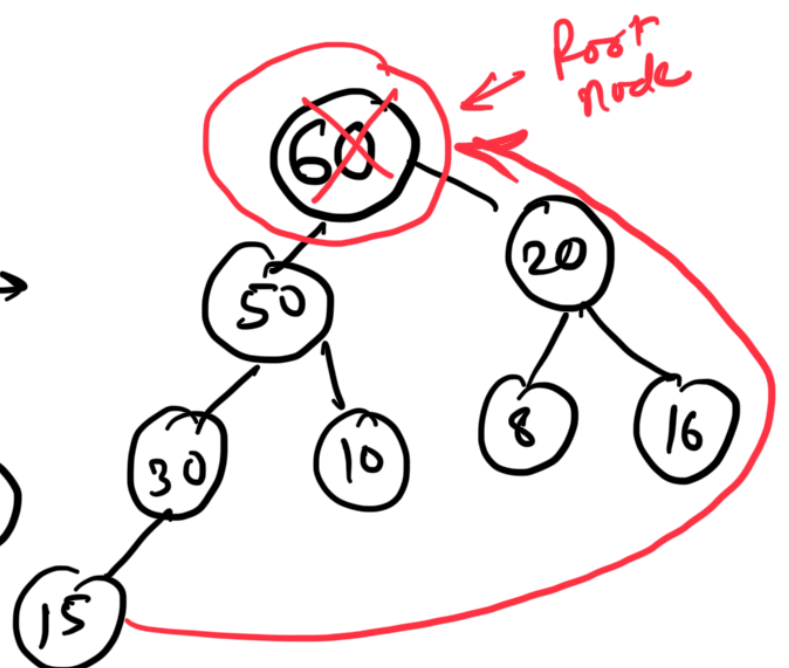
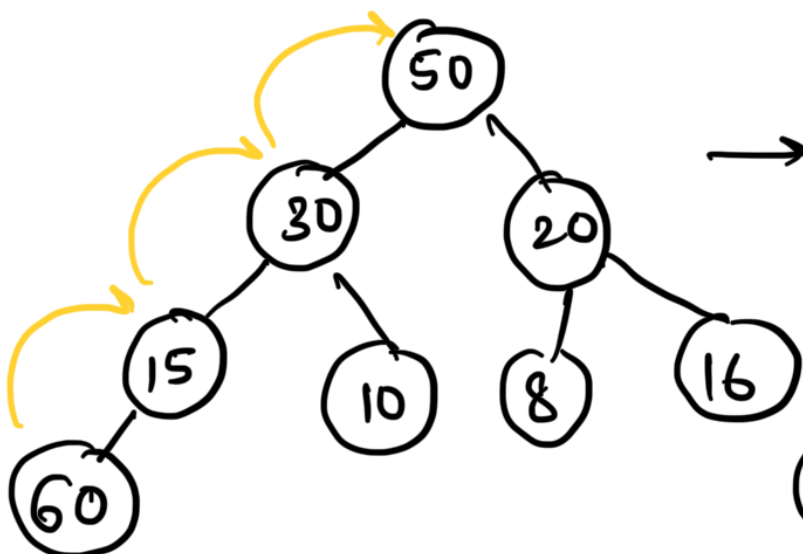
$$\text{Parent} \rightarrow \left\lfloor \frac{i}{2} \right\rfloor$$

$$LC \Rightarrow 2(1) = 2$$

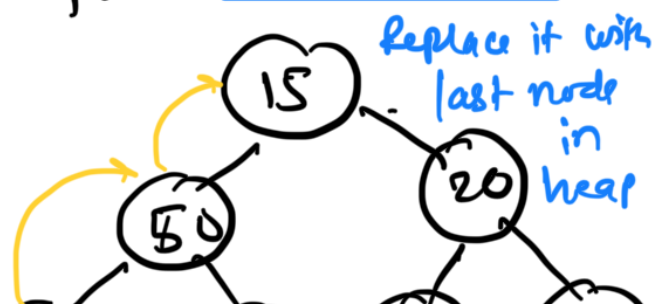
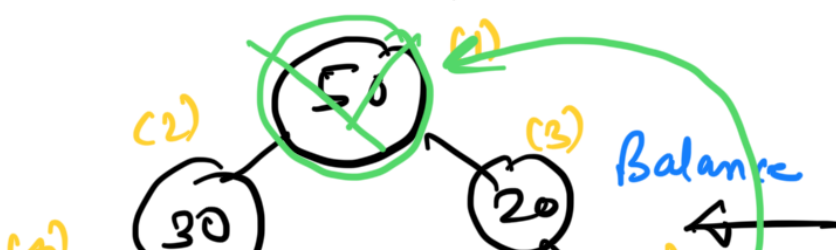
$$RC = 2(1) + 1 = 3$$



Deletion: - Max Heap

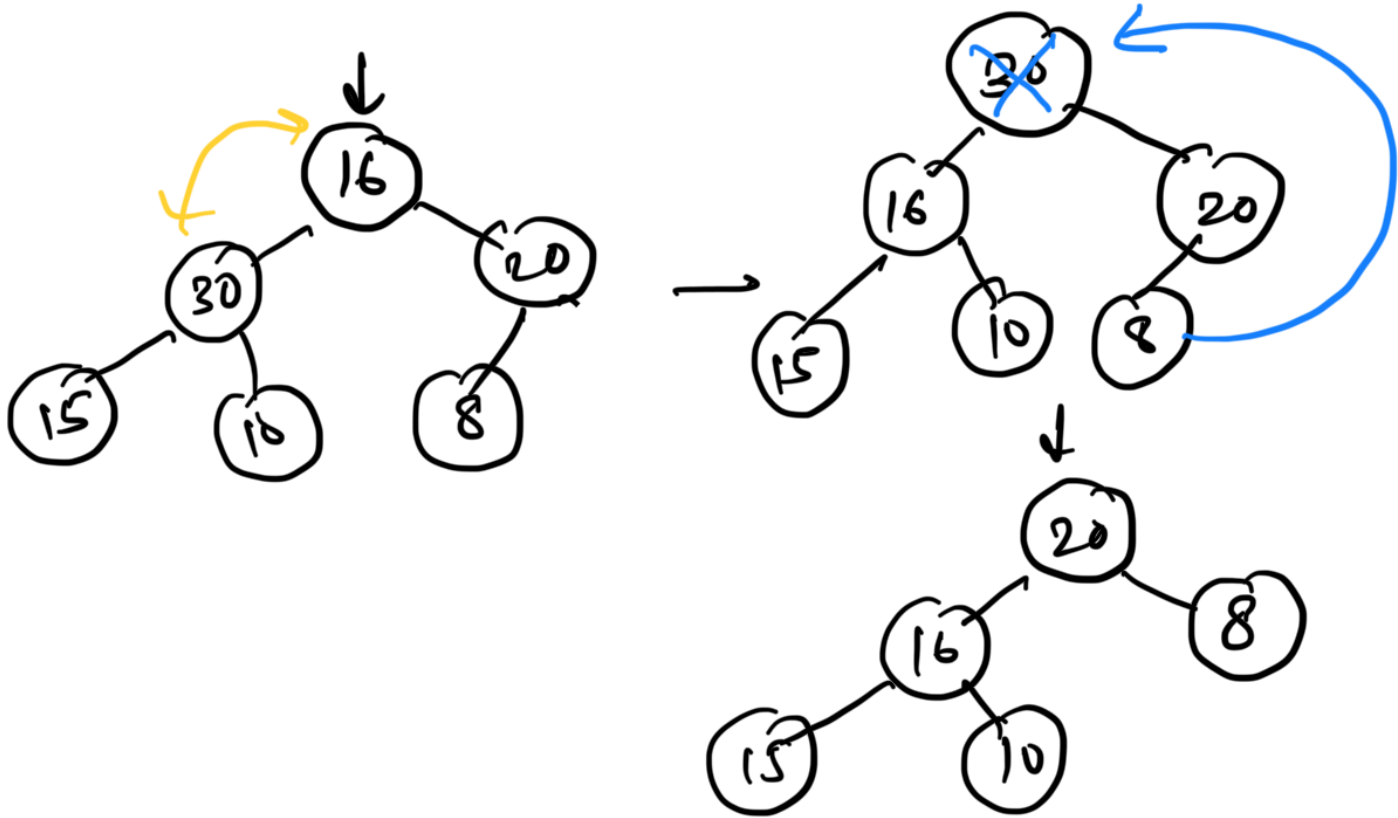


Max: Heap: Deletion is from Root node





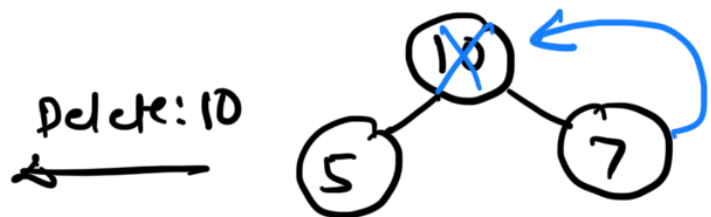
Deletion - 50



30	10	15	5	7
----	----	----	---	---

15	10	7	5	30
----	----	---	---	----

↓ Delete: 15



7	5	10	15	30
---	---	----	----	----

10	5	7	15	30
----	---	---	----	----

↓ Delete: 7



Delete: 5

X

5	7	10	15	30
---	---	----	----	----

5	7	10	15	30
---	---	----	----	----



4

Heap sort
Ascending